

Crop Planning Guide

2021

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Saskatchewan

Crop Planning Guide 2021

Each year, the Ministry of Agriculture develops the Crop Planning Guide to help producers estimate the costs and returns of producing common crops.

The detailed calculations in this guide are based on the inputs and returns associated with attaining a target yield at the 80th percentile for each soil zone. Actual costs and yields on each farm will differ due to the condition and type of equipment, the selection of crop protection products and other inputs, agronomic practices, soil class and weather conditions. Producers are expected to set their own target yields and costs. A downloadable spreadsheet is available on saskatchewan.ca/crops by searching for Crop Planning Guide.

The general agronomic and economic assumptions that apply to all crops remain consistent with previous years. Each crop has its own page with agronomic information and a blank column for producers to input their own costs and revenues. This guide also includes a yield sensitivity analysis that summarizes returns that can be expected if an operation attains an average yield, rather than the target yields used in the detailed calculations. This is intended to help producers baseline their yields and to assess any economic benefits of increasing their inputs and management efforts.

Ministry regional specialists are also available to provide advice about appropriate agronomic practices and cost assumptions.

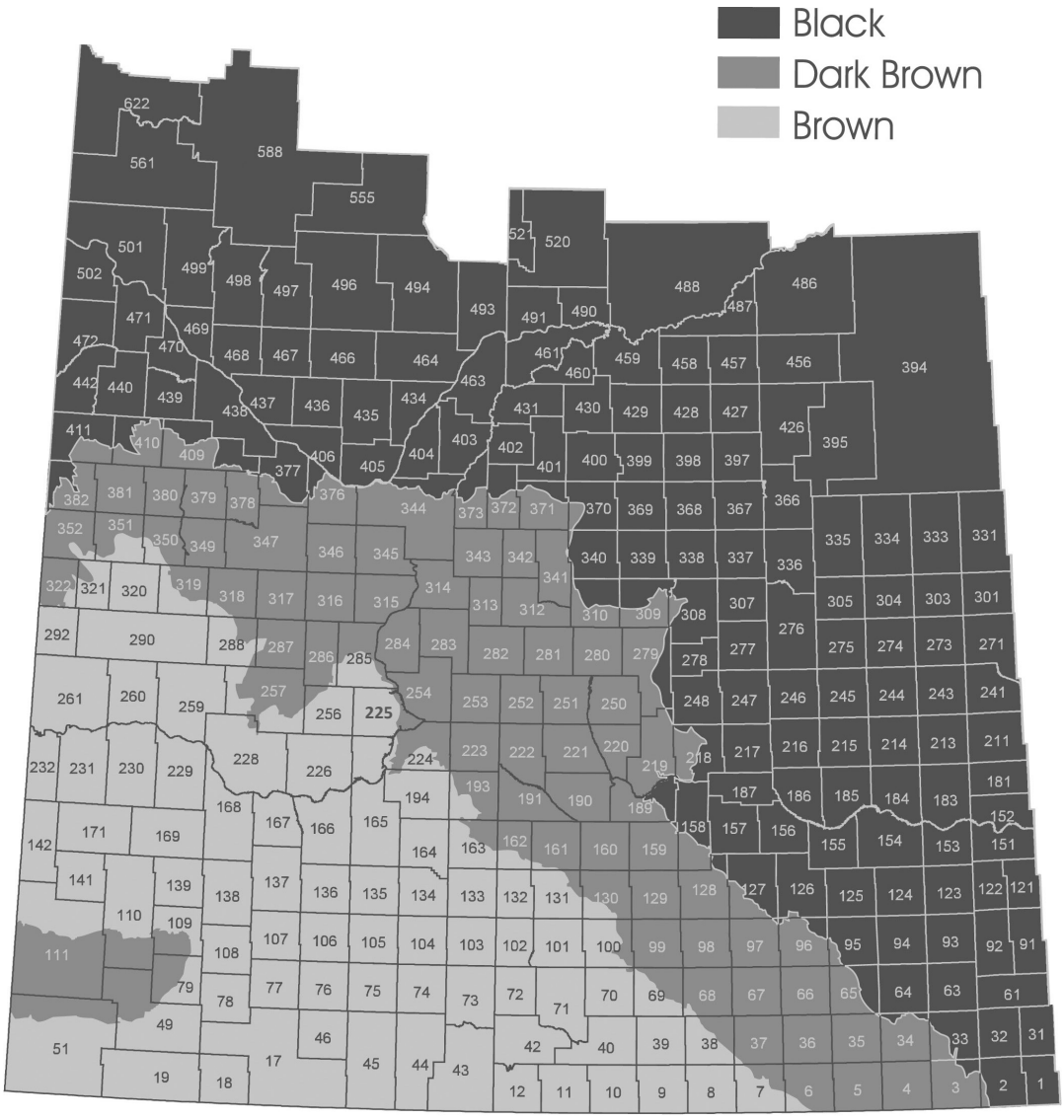


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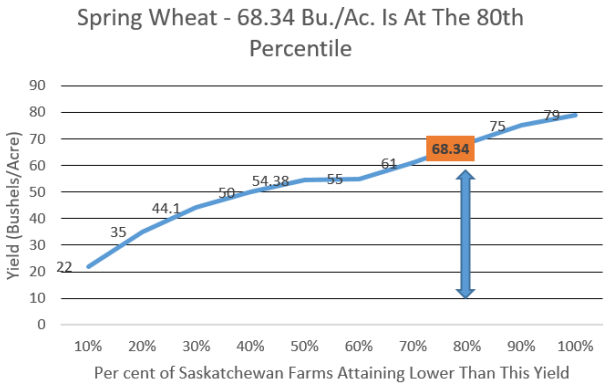
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General Assumptions for All Soil Zones

1. **Crop prices** are the average annual crop year farm gate price adopted from Agriculture and Agri-Food Canada's 2020-21 winter farm income forecast or from reports of local grain buyers. The farm gate price represents the actual payment received by farmers. This includes crops sold through forward contracts and at spot prices. Crop pricing information can become outdated quickly, so producers should adjust these figures as needed.

2. **Targeted crop yields** represent the five-year average of the 80th percentile of production for each crop in each soil zone. That is, for each of the past five years the point where 80 per cent of producers would have attained a lower yield for that crop is determined (see the image below for an example). That value for each of the five years is then averaged. The calculation uses producer data submitted to Saskatchewan Crop Insurance Corporation. These target yields reflect a higher level of management, improvements in plant genetics and higher input utilization. Producers should adjust the target yield to meet their goals and management style.



3. **Seeding rates** are determined by seed size, expected mortality, germination and desired plant population.

4. **Seed costs** are based on the use of certified seed and low disturbance direct seeding.

5. **Variety selection** should be made to best suit the agro-climatic conditions. More information can be found by searching for Varieties of Grain Crops at saskatchewan.ca/crops.

6. **Fertilizer** needs are highly variable and must be adjusted to meet conditions. The calculations in this guide are based on prices from a selection of dealers throughout the province and the estimated amount of nutrients removed from the soil in order for the crop to attain the target yield. These are not recommended application rates for specific operations. The ministry encourages producers to soil test on a consistent basis in order to measure soil fertility and calculate the total crop nutrient application required to achieve targeted crop yields. This is consistent with 4R Nutrient Stewardship management practices. For pulse crops, producers should focus on applying the correct inoculant. It is recommended to follow the guidelines for safe rates of fertilizer application with the seed when determining the right rate of phosphorous. Producers are reminded to adjust both the volume of nutrients applied and the price of each nutrient.

7. **Crop protection** efforts must be adjusted to meet each producer's conditions. The assumptions in this guide cover common applications to demonstrate potential costs. These are not recommended crop protection applications for specific operations. Producers must cost their individual responses to weed, insect and disease pressures. Please refer to the Guide to Crop Protection available at saskatchewan.ca/crops.

   The costs of crop protection products are calculated using the full registered rate of application and suggested retail prices from a selection of dealers across the province. Prices can vary significantly by vendor. Refer to the Guide to Crop Protection for registered pest control products.

Insect and disease control efforts will be aided by extended crop rotations, which reduce yield losses due to disease. Extended crop rotations also help ensure that management tools, such as resistant varieties and fungicides, remain effective by reducing selection pressure on the pathogen population. This guide assumes that commonly encountered insects or crop diseases are controlled through the use of the appropriate product given the crop and insect or disease combinations.

Weed control efforts presented reflect the practice of herbicide layering as much as possible. Herbicide layering helps prevent and manage herbicide-resistant weed populations. Layering may involve the use of two or more modes of action for control of some weeds. The timing and number of applications used to estimate herbicide costs are indicated in a chart on each crop's page. Below are the descriptions to the applications in the chart:

- **Pre-Harvest:** Pre-harvest glyphosate treatment to the previous crop. The benefit of perennial weed control from this application accrues to the crop that is planted after the application.

- **Fall Application:** Post-harvest fall application. Typically for winter annual weeds. The benefit of this application will also accrue to the crop that is planted after the application.

- **Pre-Seed:** Pre-seed burn off that replaces tillage in a low disturbance direct seeding system. There are two windows of application presented in the chart which represent one or two herbicides used at the pre-seed timing where the primary activity is on emerged weeds. Soil-active herbicides that may be mixed with the burn off applications are treated separately below. Typically, the first application will be glyphosate. When glyphosate is applied alone, it is assumed that this is done at 360 grams of acid equivalent (active) per acre. If a second application is indicated, this will be a tank mix partnered with glyphosate for burn off purposes and has limited residual

impact, if any. In a tank mix, the rate of glyphosate is assumed to be 180 grams acid equivalent (active) per acre.

- **Soil Application:** Soil-active herbicide that provides residual soil activity for control of emerging weeds beyond emergence and into the crop growth period. These are typically added for herbicide resistance management but also to contribute to increased crop yields by eliminating early weeds. Rates of some soil-active herbicides are adjusted for typical organic matter levels in the different soil zones.

- **In-Crop:** In-crop foliar application opportunities are provided for up to three herbicides. These are applied either in a mix or as separate sequential applications where tank mixing is not compatible. There are three windows of application allotted as indicated by each column in the chart.

- **Desiccation:** Harvest aid application prior to harvest. This is done for the purpose of rapid dry down of the crop to facilitate timely harvest. This does not include glyphosate, which may be included as a mix with some harvest aid options.

8. **Machinery operating** costs include fuel usage and repair. Fuel costs are based on estimated fuel consumptions for the various farming operations with diesel fuel priced at $0.906/litre. Machinery repair rates are based on the Ministry's 2020-21 Custom Rate and Rental Guide and are set at 2.6 per cent of the yearly machinery investment cost.

9. **Custom work and hired labour** is made up of costs for custom farm operations, such as custom trucking and custom spraying. Skilled labour is assumed to be $26.40 per hour for 2021.

10. **Crop insurance premiums** are the five-year average of the premiums paid by producers who attain the targeted yield for the soil zone. The premiums used in this guide do not reflect actual producers' costs given surcharges and taxes.

11. **Utilities** include the costs of electricity, natural gas, water and telephone expenses based on the standard farm business rates of major utility providers.

12. **Interest on variable expenses** is calculated using a rate of 2.95 per cent on all variable expenses. The interest is applied for eight months for all crops except hybrid fall rye and winter wheat. For these two crops, the interest costs are calculated for 18 months.

13. **Building repair** rates are 2.3 per cent of building investment per acre.

14. **Business overhead** is made up of legal, accounting, insurance, licences and miscellaneous. Business overhead costs are indexed by applying the farm input price index to Statistics Canada's 2016 Census of Agriculture.

15. **Machinery investment** costs are calculated by applying a interest rate to 75 per cent of machinery investment and an opportunity cost to the remaining 25 per cent that would have been provided as a down payment on machinery. This is a new approach to calculating these costs and is consistent with the Ministry of Agriculture's Custom Rate and Rental Guide assumptions. An interest rate of 5 per cent and an opportunity cost rate of 1.5 per cent are utilized. These calculations are applied to the average investment in machinery by soil zone. Based on Statistics Canada data, it is estimated that a brown soil zone farm has $346.25 per cultivated acre invested in machinery, a dark brown soil zone farm has $390.45 per cultivated acre invested and a black soil zone farm has $441.91 invested per cultivated acre. Machinery costs are substantial and these are average rates per soil zone. Producers are reminded to adjust these figures to meet their unique circumstances.

16. **Machinery depreciation** is calculated using a straight-line formula at 10.7 per cent annual depreciation rate.

17. **Building investment** is calculated as 2.19 per cent interest on annual building investment. Based on data provided by Statistics Canada, it is estimated that a brown soil zone farm has $21 per cultivated acre invested in buildings, a dark brown soil zone farm has $28 per cultivated acre invested and a black soil zone farm has $38 per cultivated acre invested.

18. **Building depreciation** is calculated at five per cent per year on a straight-line basis of building investment.

19. **Land investment** cost is calculated as 2.19 per cent interest on an investment of $1,633.3 per cultivated acre in the brown soil zone, $1,840.79 per cultivated acre in the dark brown soil zone and $1,757.80 per cultivated acre in the black soil zone. This year's calculations show lower costs for land investment because borrowing conditions have become more favorable. Producers should adjust this figure to reflect payments they make towards land. This is where any land rental rates should be accounted for.

20. **Labour and management** refers to owner/operator labour and management and is not included in these estimates.

2021 Feed Barley

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		78.1	85.4	96.5
Estimated Farm Gate Price ($/bu.) (B)		4.37	4.37	4.37
Estimated Gross Revenue ($/ac.) (AxB)=(C)		341.50	373.42	421.96
Expenses Per Acre				
Variable Expenses/Acre				
Seed		17.41	19.19	21.76
- Seed Treatments/Inoculants		8.05	8.87	10.06
Fertilizer - Nitrogen		39.71	43.53	48.79
- Phosphorus (P_2O_5)		17.09	18.51	21.36
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		20.82	20.82	20.82
- Insecticides		26.91	26.91	26.91
- Fungicides		0.00	0.00	19.10
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.80	21.05	21.05
Crop Insurance Premium		5.84	5.20	4.90
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		3.56	3.81	4.53
Total Variable Expenses (D)		**184.74**	**197.68**	**234.91**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**280.08**	**307.21**	**354.48**
Return Per Acre				
Return Over Variable Expenses (C-D)		156.76	175.74	187.05
Return Over Total Expenses (C-G)		61.42	66.21	67.48
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		42.25	45.21	53.72
To Cover Total Expenses		64.05	70.26	81.07
Break Even Price ($/bu.)				
To Cover Variable Expenses		2.37	2.31	2.43
To Cover Total Expenses		3.59	3.60	3.67

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	61.55	58.33	74.87
Return Over Variable Expenses	84.39	57.37	92.47
Return Over Total Expenses	-10.95	-52.16	-27.10

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: Seeding rates are based on a target plant stand in plants per square foot. This guide assumes a producer will seed to achieve 25 plants per square foot in the black soil zone, 22 in the dark brown soil zone and 20 in the brown soil zone, with a thousand kernel weight of 45 grams and 85 per cent emergence.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 102 lb./ac. N and 45 lb./ac. P_2O_5 for the black soil zone, 91 lb./ac. N and 39 lb./ac. P_2O_5 for the dark brown soil zone and 83 lb./ac. N and 36 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weeds to manage herbicide resistance. Barley is a very competitive crop that will suppress growth of weeds. Feed barley markets may be more tolerant of weed escapes than malt barley.

Crop Protection

Insect control: Cutworms, aphids, thrips, mites, grasshoppers, armyworm, slugs and wireworms might require control. Seed treatments are available for wireworm control.

Disease control: Cereal crops can be affected by both leaf diseases and fusarium head blight (FHB). When disease pressure is moderate, a single application at FHB timing can be effective in managing both types of diseases. This estimation includes the cost of a single fungicide application in the black soil zone. When disease pressure is high, an additional fungicide application for leaf diseases may be required. Fungicide applications should be made based on field history and disease risk during the growing season.

Weed control: Because barley is competitive, growers can often reduce the number of herbicide applications from those listed. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓		✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Malt Barley

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		63.9	69.9	78.3
Estimated Farm Gate Price ($/bu.) (B)		5.20	5.20	5.20
Estimated Gross Revenue ($/ac.) (AxB)=(C)		332.28	363.48	407.16
Expenses Per Acre				
Variable Expenses/Acre				
Seed		24.00	26.26	29.88
- Seed Treatments/Inoculants		9.70	10.61	12.07
Fertilizer - Nitrogen		32.53	35.40	39.71
- Phosphorus (P_2O_5)		14.24	15.19	17.09
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		59.72	59.72	59.72
- Insecticides		26.91	26.91	26.91
- Fungicides		19.10	19.10	19.10
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.80	21.05	21.05
Crop Insurance Premium		4.79	4.26	3.98
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		4.65	4.88	5.21
Total Variable Expenses (D)		**240.98**	**253.17**	**270.35**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**336.32**	**362.70**	**389.92**
Return Per Acre				
Return Over Variable Expenses (C-D)		91.30	110.31	136.81
Return Over Total Expenses (C-G)		-4.04	0.78	17.24
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		46.34	48.69	51.99
To Cover Total Expenses		64.68	69.75	74.99
Break Even Price ($/bu.)				
To Cover Variable Expenses		3.77	3.62	3.45
To Cover Total Expenses		5.26	5.19	4.98

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	50.41	47.77	60.79
Return Over Variable Expenses	21.15	-4.77	45.76
Return Over Total Expenses	-74.19	-114.30	-73.81

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.Calculations provided are for the 80th percentile of production.

Seeding: Seeding rates are based on a target plant stand in plants per square foot. This guide assumes a producer will seed to achieve 25 plants per square foot in the black soil zone, 22 in the dark brown soil zone and 20 brown soil zone, with a thousand kernel weight of 45 grams and 85 per cent emergence.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 83 lb./ac. N and 36 lb./ac. P_2O_5 for the black soil zone, 74 lb./ac. N and 32 lb./ac. P_2O_5 for the dark brown soil zone and 68 lb./ac. N and 30 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weeds to manage herbicide resistance. Barley is a very competitive crop that will suppress growth of weeds.

Crop Protection

Insect control: Cutworms, aphids, thrips, mites, grasshoppers, armyworm, slugs and wireworms might require control. Seed treatments are available for wireworm control.

Disease control: Cereal crops can be affected by both leaf diseases and fusarium head blight (FHB). When disease pressure is moderate, a single application at FHB timing can be effective in managing both types of diseases. This estimation includes the cost of a single fungicide application. When disease pressure is high, an additional fungicide application for leaf diseases may be required. Fungicide applications should be made based on field history and disease risk during the growing season.

Weed control: Because barley is competitive, growers can often reduce the number of herbicide applications from those listed. A soil-applied herbicide was used to manage Group 1 resistant wild oats. Refer to the Guide to Crop Protection available at saskatchewan.ca/crops for more information about Group 1 resistance. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Corn

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		76.8	76.8	117.7
Estimated Farm Gate Price ($/bu.) (B)		4.85	4.85	4.85
Estimated Gross Revenue ($/ac.) (AxB)=(C)		372.32	372.48	570.85
Expenses Per Acre				
Variable Expenses/Acre				
Seed		96.30	96.30	96.30
- Seed Treatments/Inoculants		0.00	0.00	0.00
Fertilizer - Nitrogen		39.23	39.23	60.28
- Phosphorus (P_2O_5)		17.56	17.56	26.58
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		49.90	38.95	49.90
- Insecticides		26.91	26.91	26.91
- Fungicides		0.00	0.00	0.00
Machinery Operating - Fuel		13.77	17.21	21.52
- Repair		11.61	12.76	14.10
Custom Work and Hired Labour		49.04	45.48	58.26
Crop Insurance Premium		6.47	7.31	10.70
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		6.18	6.02	7.27
Total Variable Expenses (D)		**320.20**	**311.97**	**376.70**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		47.79	52.52	58.03
Building Depreciation		1.05	1.40	1.90
Machinery Investment		18.42	20.25	22.37
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**110.23**	**124.42**	**134.46**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**430.43**	**436.38**	**511.17**
Return Per Acre				
Return Over Variable Expenses (C-D)		52.12	60.51	194.15
Return Over Total Expenses (C-G)		-58.11	-63.90	59.68
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		66.02	64.32	77.67
To Cover Total Expenses		88.75	89.98	105.39
Break Even Price($/bu.)				
To Cover Variable Expenses		4.17	4.06	3.20
To Cover Total Expenses		5.61	5.68	4.34

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	67.71	65.35	91.73
Return Over Variable Expenses	8.19	4.98	68.19
Return Over Total Expenses	-102.04	-119.43	-66.28

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Variety Selection: Corn varieties are not listed in the Varieties of Grain Crops found on saskatchewan.ca/crops. Please contact your retailer for more information.

Seeding: A plant population of 30,000 plants/ac is used for all three soil zones.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 126 lb./ac. N and 56 lb./ac. P_2O_5 for black soil zone and 82 lb./ac. N and 37 lb./ac. P_2O_5 in brown and dark brown soil zones. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Extended crop rotations can be used to reduce disease pressure by allowing infected residue to decompose between host crops. This is particularly important for residue-borne diseases caused by bacteria, such as Goss's Wilt, as fungicides will not protect against this disease. Corn is not competitive with weeds. Some herbicide choices in corn can significantly restrict cropping options the following year.

Crop Protection

Insect control: Cutworms, wireworms, seedcorn maggot, corn rootworm, aphids, spider mite, grasshoppers, European corn borer, corn earworm and armyworms might require control. Seed treatments are available for wireworm and seedcorn maggot control. Varieties resistant to European corn borer, corn earworm, and corn rootworm are available.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Corn must be kept free of weeds until 10 leaf tips are visible to prevent significant yield losses. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓	✓	✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Hybrid Fall Rye

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		50.8	66.9	78.3
Estimated Farm Gate Price ($/bu.) (B)		5.10	5.10	5.10
Estimated Gross Revenue ($/ac.) (AxB)=(C)		259.08	341.19	399.33
Expenses Per Acre				
Variable Expenses/Acre				
Seed		67.20	67.20	67.20
- Seed Treatments/Inoculants		0.00	0.00	0.00
Fertilizer - Nitrogen		28.22	37.31	43.53
- Phosphorus (P_2O_5)		11.87	15.66	18.04
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		22.30	22.30	22.30
- Insecticides		0.00	0.00	0.00
- Fungicides		0.00	0.00	0.00
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.80	20.80	21.05
Crop Insurance Premium		8.50	5.76	5.80
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		8.12	8.80	9.45
Total Variable Expenses (D)		**191.56**	**207.63**	**223.00**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**286.90**	**317.16**	**342.57**
Return Per Acre				
Return Over Variable Expenses (C-D)		67.52	133.56	176.33
Return Over Total Expenses (C-G)		-27.82	24.03	56.76
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		37.56	40.71	43.72
To Cover Total Expenses		56.26	62.19	67.17
Break Even Price ($/bu.)				
To Cover Variable Expenses		3.77	3.10	2.85
To Cover Total Expenses		5.65	4.74	4.38
Yield Sensitivity (same expenses, but average yield)				
Provincial Average Yield (bu./ac.)		38.97	38.97	31.10
Return Over Variable Expenses		7.19	-8.88	-64.39
Return Over Total Expenses		-88.15	-118.41	-183.96

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: Certified seed is needed every year for hybrids. A seeding rate of 0.8 units/ac. is used for all soil zones. One unit is equal to one million viable seeds.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 91 lb./ac N and 38 lb./ac. P_2O_5 for the black soil zone, 78 lb./ac. N and 33 lb./ac. P_2O_5 for the dark brown soil zone and 59 lb./ac. N and 25 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weeds to manage herbicide resistance. Fall rye is a very competitive crop that will suppress growth of spring germinating weeds.

Crop Protection

Insect control: Cutworms, aphids, thrips, mites, grasshoppers, armyworm, slugs and wireworms might require control. Seed treatments are available for wireworm control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Rye has very few herbicide options. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓		✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Oats

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		81.1	110.2	145.9
Estimated Farm Gate Price ($/bu.) (B)		3.33	3.33	3.33
Estimated Gross Revenue ($/ac.) (AxB)=(C)		270.06	366.97	485.81
Expenses Per Acre				
Variable Expenses/Acre				
Seed		31.43	31.43	31.43
- Seed Treatments/Inoculants		12.35	12.35	12.35
Fertilizer - Nitrogen		26.31	35.88	47.36
- Phosphorus (P_2O_5)		10.92	14.72	19.46
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		24.32	24.32	24.32
- Insecticides		26.91	26.91	26.91
- Fungicides		0.00	0.00	19.10
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.80	21.05	21.05
Crop Insurance Premium		8.33	7.88	7.14
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		3.66	4.02	4.81
Total Variable Expenses (D)		**189.58**	**208.34**	**249.56**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**284.92**	**317.87**	**369.14**
Return Per Acre				
Return Over Variable Expenses (C-D)		80.48	158.63	236.25
Return Over Total Expenses (C-G)		-14.86	49.10	116.67
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		56.93	62.57	74.94
To Cover Total Expenses		85.56	95.46	110.85
Break Even Price ($/bu.)				
To Cover Variable Expenses		2.34	1.89	1.71
To Cover Total Expenses		3.51	2.88	2.53

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	47.33	66.14	100.51
Return Over Variable Expenses	-31.97	11.91	85.14
Return Over Total Expenses	-127.31	-97.62	-34.44

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seed rate of 135 lb./ac is used for all soil zones.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 99 lb./ac. N and 41 lb./ac. P_2O_5 for the black soil zone, 75 lb./ac. N and 31 lb./ac. P_2O_5 for the dark brown soil zone and 55 lb./ac. N and 23 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure as well as suppress weeds to manage herbicide resistance. Oat is a very competitive crop that will suppress growth of spring germinating weeds. Wild oats cannot be controlled in tame oat with herbicides.

Crop Protection

Insect control: Cutworms, aphids, thrips, mites, grasshoppers, armyworm, slugs and wireworms might require control. Seed treatments are available for wireworm control.

Disease control: Leaf diseases may result in yield losses in oat crops. Fungicide application can be used to protect leaf tissue from disease infection. This estimation includes the cost of a single fungicide application in the black soil zone. Fungicide application should be based on disease pressure in the field.

Weed control: Because oats are very competitive, growers can often reduce the number of herbicide applications from those listed. Some buyers of milling oats do not allow use of pre-harvest glyphosate in their contracts. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
	✓	✓	✓		✓			

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Durum Wheat

Economics

	My Farm	Brown	Dark Brown
Revenue Per Acre			
Target Yield (bu./ac.) (A)		54.0	61.7
Estimated Farm Gate Price ($/bu.) (B)		7.50	7.50
Estimated Gross Revenue ($/ac.) (AxB)=(C)		405.10	462.97
Expenses Per Acre			
Variable Expenses/Acre			
Seed		26.83	29.64
- Seed Treatments/Inoculants		9.60	10.61
Fertilizer - Nitrogen		42.58	48.79
- Phosphorus (P_2O_5)		16.61	18.99
- Sulphur and Other		0.00	0.00
Plant Protection - Herbicides		16.40	16.40
- Insecticides		26.91	26.91
- Fungicides		19.10	19.10
Machinery Operating - Fuel		12.32	15.40
- Repair		9.00	10.15
Custom Work and Hired Labour		22.30	22.05
Crop Insurance Premium		6.10	7.94
Utilities and Miscellaneous		3.23	4.23
Interest on Variable Expenses		4.15	4.53
Total Variable Expenses (D)		**215.13**	**234.75**
Other Expenses/Acre			
Building Repair		0.49	0.65
Property Taxes		3.97	5.19
Business Overhead		2.28	3.48
Machinery Depreciation		37.05	41.78
Building Depreciation		1.05	1.40
Machinery Investment		14.28	16.11
Building Investment		0.46	0.61
Land Investment		35.77	40.31
Total Other Expenses (E)		**95.34**	**109.53**
Labour and Management* (F)			
Total Expenses (D+E+F)=(G)		**310.47**	**344.28**
Return Per Acre			
Return Over Variable Expenses (C-D)		189.97	228.22
Return Over Total Expenses (C-G)		94.63	118.69
Break Even Yield (bu./ac.)			
To Cover Variable Expenses		28.68	31.30
To Cover Total Expenses		41.40	45.90
Break Even Price($/bu.)			
To Cover Variable Expenses		3.98	3.80
To Cover Total Expenses		5.75	5.58

Yield Sensitivity (same expenses, but average yield)		
Provincial Average Yield (bu./ac.)	42.26	43.73
Return Over Variable Expenses	101.82	93.23
Return Over Total Expenses	6.48	-16.30

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: Seeding rates are based on a target plant stand in plants per square foot. This guide assumes a producer will seed to achieve 22 plants per square foot in dark brown soil zone and 20 in the brown soil zone, with a thousand kernel weight of 42.1 grams and 85 per cent emergence. Durum is recommended in the brown and dark brown soil zones.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 102 lb./ac. N and 40 lb./ac. P_2O_5 for the dark brown soil zone and 89 lb./ac. N and 35 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weed growth. Like all cereals, durum is relatively competitive crop against weeds.

Crop Protection

Insect control: Wheat midge, cutworms, aphids, thrips, mite, grasshoppers, armyworms, slugs, wheat stem sawfly and wireworms might require control. Varietal blends with resistance are available if heavy wheat midge pressures are anticipated. Wheat stem sawfly resistant varieties are available. Seed treatments are available for wireworm control. An insecticide application to control wheat midge is assumed, but in practice should be made based on scouting to determine economic risk. No insecticide applications for wheat midge would be required for midge tolerant varieties.

Disease control: Cereal crops can be affected by both leaf diseases and fusarium head blight (FHB). When disease pressure is moderate, a single application at FHB timing can be effective in managing both types of diseases. This estimation includes the cost of a single fungicide application. When disease pressure is high an additional fungicide application for leaf diseases might be required. Fungicide applications should be made based on field history and disease risk during the growing season.

Weed control: Durum lacks many options for soil-applied herbicides for herbicide layering programs. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓		✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Spring Wheat

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		52.9	61.4	68.3
Estimated Farm Gate Price ($/bu.) (B)		6.04	6.04	6.04
Estimated Gross Revenue ($/ac.) (AxB)=(C)		319.58	370.63	412.80
Expenses Per Acre				
Variable Expenses/Acre				
Seed		18.45	20.38	23.17
- Seed Treatments/Inoculants		7.87	8.69	9.88
Fertilizer - Nitrogen		41.62	48.32	54.06
- Phosphorus (P_2O_5)		16.14	18.99	20.89
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		61.70	61.70	62.20
- Insecticides		26.91	26.91	26.91
- Fungicides		19.10	19.10	19.10
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		22.30	22.05	23.05
Crop Insurance Premium		4.46	4.18	3.87
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		4.78	5.12	5.48
Total Variable Expenses (D)		**247.88**	**265.22**	**284.23**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**343.22**	**374.75**	**403.80**
Return Per Acre				
Return Over Variable Expenses (C-D)		71.70	105.41	128.57
Return Over Total Expenses (C-G)		-23.64	-4.12	9.00
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		41.04	43.91	47.06
To Cover Total Expenses		56.82	62.05	66.85
Break Even Price ($/bu.)				
To Cover Variable Expenses		4.68	4.32	4.16
To Cover Total Expenses		6.49	6.11	5.91

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	40.42	41.15	54.38
Return Over Variable Expenses	-3.74	-16.67	44.23
Return Over Total Expenses	-99.08	-126.20	-75.34

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: Seeding rates are based on a target plant stand in plants per square foot. This guide assumes a producer will seed to achieve 25 plants per square foot in the black soil zone, 22 in the dark brown soil zone and 20 in the brown soil zone, with a thousand kernel weight of 34.5 grams and 85 per cent emergence.

Fertilization: Fertility costs are based on nutrient removal rates given the target yield. These are: 113 lb./ac. N and 44 lb./ac. P_2O_5 for the black soil zone, 101 lb./ac. N and 40 lb./ac. P_2O_5 for the dark brown soil zone and 87 lb./ac. N and 34 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Rotation plays an important role in the suppression of weed growth. A break between cereal crops will help reduce disease pressure by allowing infested crop residue to decompose.

Crop Protection

Insect control: Wheat midge, cutworms, aphids, thrips, mites, grasshoppers, armyworms, slugs, wheat stem sawfly and wireworms might require control. An insecticide application to control wheat midge is assumed, but in practice should be made based on scouting to determine economic risk. No insecticide application for wheat midge would be required for midge tolerant varieties. Please refer to the Guide to Crop Protection available at saskatchewan.ca/crops for registered pest control products for specific pests.

Disease control: Cereal crops can be affected by both leaf diseases and fusarium head blight (FHB). When disease pressure is moderate, a single application at FHB timing can be effective in managing both types of diseases. This estimation includes the cost of a single fungicide application. When disease pressure is high, an additional fungicide application for leaf diseases may be required. Fungicide applications should be made based on field history and disease risk during the growing season.

Weed control: Spring wheat has many herbicide options to choose from. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓	✓	✓			

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Winter Wheat

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		59.5	60.3	71.3
Estimated Farm Gate Price ($/bu.) (B)		5.83	5.83	5.83
Estimated Gross Revenue ($/ac.) (AxB)=(C)		346.89	351.55	415.68
Expenses Per Acre				
Variable Expenses/Acre				
Seed		28.13	28.13	28.13
- Seed Treatments/Inoculants		8.69	8.69	8.69
Fertilizer - Nitrogen		32.53	33.01	38.75
- Phosphorus (P_2O_5)		15.66	16.14	18.99
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		50.94	50.94	50.94
- Insecticides		12.68	12.68	12.68
- Fungicides		0.00	19.10	19.10
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		22.30	22.05	23.05
Crop Insurance Premium		8.34	8.37	6.82
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		9.02	10.13	10.74
Total Variable Expenses (D)		**212.84**	**239.02**	**253.51**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**308.18**	**348.55**	**373.08**
Return Per Acre				
Return Over Variable Expenses (C-D)		134.05	112.53	162.17
Return Over Total Expenses (C-G)		38.71	3.00	42.60
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		36.51	41.00	43.48
To Cover Total Expenses		52.86	59.79	63.99
Break Even Price ($/bu.)				
To Cover Variable Expenses		3.58	3.96	3.56
To Cover Total Expenses		5.18	5.78	5.23
Yield Sensitivity (same expenses, but average yield)				
Provincial Average Yield (bu./ac.)		41.89	44.83	44.46
Return Over Variable Expenses		31.38	22.34	5.69
Return Over Total Expenses		-63.96	-87.19	-113.88

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: Seeding rates are based on a target plant stand in plants per square foot. This guide assumes a producer will seed to achieve 30 plants per square foot in the black soil zone, 28 in the dark brown soil zone and 25 in the brown soil zone, with a thousand kernel weight of 36 grams and 85 per cent emergence.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 81 lb./ac. N and 40 lb./ac. P_2O_5 for the black soil zone, 69 lb./ac. N and 34 lb./ac. P_2O_5 for the dark brown soil zone and 68 lb./ac. N and 33 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weeds to manage herbicide resistance. Winter wheat is a very competitive crop that will suppress growth of spring germinating weeds.

Crop Protection

Insect control: Wheat midge, cutworms, aphids, thrips, mites, grasshoppers, armyworms, slugs and wireworms might require control. Seed treatments are available for wireworm control.

Disease control: Fungicide applications in winter wheat typically target leaf diseases. This estimation includes the cost of a single fungicide application for leaf diseases in the black and dark brown soil zone. Fungicide applications should be made based on field history and disease risk during the growing season. Winter wheat can be affected by both leaf diseases and fusarium head blight (FHB). However, winter wheat crops typically pass the susceptible growth stage when conditions favor FHB development.

Weed control: Winter wheat is prone to infestation with winter annual weeds, particularly downy and Japanese brome. Herbicide choices were made with this weed in mind, but should be adjusted on individual farms based on the weeds present. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Canola

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		48.06	52.00	56.00
Estimated Farm Gate Price ($/bu.) (B)		11.25	11.25	11.25
Estimated Gross Revenue ($/ac.) (AxB)=(C)		540.68	585.00	630.00
Expenses Per Acre				
Variable Expenses/Acre				
Seed		70.55	70.55	70.55
- Seed Treatments/Inoculants		9.00	9.00	9.00
Fertilizer - Nitrogen		48.79	52.62	56.45
- Phosphorus (P_2O_5)		26.11	28.01	30.38
- Sulphur and Other		5.11	5.75	6.13
Plant Protection - Herbicides		57.58	57.58	64.37
- Insecticides		21.86	21.86	21.86
- Fungicides		0.00	28.13	28.13
Machinery Operating - Fuel		13.05	16.31	20.39
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		21.05	21.05	21.05
Crop Insurance Premium		10.40	8.16	8.60
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		5.82	6.56	6.95
Total Variable Expenses (D)		**301.54**	**339.96**	**360.23**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**396.89**	**449.49**	**479.80**
Return Per Acre				
Return Over Variable Expenses (C-D)		239.14	245.04	269.77
Return Over Total Expenses (C-G)		143.79	135.51	150.20
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		26.80	30.22	32.02
To Cover Total Expenses		35.28	39.95	42.65
Break Even Price ($/bu.)				
To Cover Variable Expenses		6.27	6.54	6.43
To Cover Total Expenses		8.26	8.64	8.57

Yield Sensitivity (same expenses, but average yield)

	Brown	Dark Brown	Black
Provincial Average Yield (bu./ac.)	28.22	48.06	40.12
Return Over Variable Expenses	15.94	200.72	91.12
Return Over Total Expenses	-79.41	91.19	-28.45

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seeding rate of 5 lb./ac. is used for each soil zone.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 118 lb./ac. N and 64 lb./ac. P_2O_5 and 19 lb./ac. S for the black soil zone, 110 lb./ ac. N and 59 lb./ac. P_2O_5 and 18 lb./ac. S for the dark brown soil zone and 102 lb./ac. N and 55 lb./ac. P_2O_5 and 16 lb./ac. S for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation will help to reduce root maggot and pressure from diseases, such as clubroot, by reducing or maintaining low pathogen levels in the field.

Crop Protection

Insect control: Flea beetles, cutworms, lygus bugs, cabbage seedpod weevil, diamondback moth, bertha armyworm, alfalfa looper, cabbage looper, and occasionally imported cabbageworm, grasshoppers and slugs might require control. Seed treatments are available for flea beetle and cutworm control.

Disease control: Sclerotinia stem rot is the main disease managed with the application of foliar fungicides. This estimation includes the cost of a single fungicide application in the dark brown and black soil zones. Disease pressure will vary from year to year and field to field and is influenced by environmental conditions. Fungicide application decisions should be made based on disease risk when the crop is susceptible to infection.

Weed control: A soil-active herbicide to reduce competition from cleavers was included in brown and dark brown soils. This was exchanged for a foliar tank mix option in the black soils. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓		✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Flax

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		30.3	34.3	37.4
Estimated Farm Gate Price ($/bu.) (B)		13.97	13.97	13.97
Estimated Gross Revenue ($/ac.) (AxB)=(C)		423.29	479.17	522.48
Expenses Per Acre				
Variable Expenses/Acre				
Seed		18.12	19.63	20.86
- Seed Treatments/Inoculants		1.80	1.95	2.10
Fertilizer - Nitrogen		33.96	38.27	41.62
- Phosphorus (P_2O_5)		9.97	11.39	12.34
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		45.84	45.84	29.29
- Insecticides		25.76	25.76	25.76
- Fungicides		0.00	28.13	28.13
Machinery Operating - Fuel		12.32	15.40	19.25
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.80	21.05	21.05
Crop Insurance Premium		9.01	7.02	6.43
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		3.73	4.50	4.39
Total Variable Expenses (D)		**193.54**	**233.33**	**227.59**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**288.88**	**342.86**	**347.17**
Return Per Acre				
Return Over Variable Expenses (C-D)		229.75	245.84	294.89
Return Over Total Expenses (C-G)		134.41	136.31	175.31
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		13.85	16.70	16.29
To Cover Total Expenses		20.68	24.54	24.85
Break Even Price ($/bu.)				
To Cover Variable Expenses		6.39	6.80	6.09
To Cover Total Expenses		9.53	10.00	9.28

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	27.56	30.31	23.23
Return Over Variable Expenses	191.47	190.10	96.93
Return Over Total Expenses	96.13	80.57	-22.65

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: Seeding rates used are 42 lb./ac. in the black soil zone, 39 lb./ac. in the dark brown soil zone and 36 lb./ac. in the brown soil zone.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 87 lb./ac. N and 26 lb./ac. P_2O_5 for the black soil zone, 80 lb./ac. N and 24 lb./ac. P_2O_5 for the dark brown soil zone and 71 lb./ac. N and 21 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure by allowing infected crop residue to decompose between susceptible crops. Flax is not competitive against weeds and very sensitive to herbicide residues in the soil.

Crop Protection

Insect control: Wireworms, cutworms, lygus bugs, potato aphid, grasshoppers, bertha armyworm, armyworm and beet webworm might require control.

Disease control: A single fungicide application for pasmo management has been included in this estimate. Early pasmo infection can result in losses of yield and quality. Fungicide application should be based on disease risk when the crop is susceptible to disease infection.

Weed control: Flax has limited herbicide options. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓	✓	✓	✓		✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Brown Mustard

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,301.00
Estimated Farm Gate Price ($/lb.) (B)		0.35
Estimated Gross Revenue ($/ac.) (AxB)=(C)		455.35
Expenses Per Acre		
Variable Expenses/Acre		
Seed		23.10
- Seed Treatments/Inoculants		0.54
Fertilizer - Nitrogen		23.92
- Phosphorus (P_2O_5)		12.82
- Sulphur and Other		4.79
Plant Protection - Herbicides		42.58
- Insecticides		8.12
- Fungicides		0.00
Machinery Operating - Fuel		19.03
- Repair		9.00
Custom Work and Hired Labour		20.30
Crop Insurance Premium		7.07
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		3.43
Total Variable Expenses (D)		**177.92**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**273.26**
Return Per Acre		
Return Over Variable Expenses (C-D)		277.43
Return Over Total Expenses (C-G)		182.09
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		508.34
To Cover Total Expenses		780.74
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.14
To Cover Total Expenses		0.21

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	815.70
Return Over Variable Expenses	107.58
Return Over Total Expenses	12.24

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 6 lb./ac. is used. The brown soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 50 lb./ac. N, 27 lb./ac. P_2O_5 and 15 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weed growth.

Crop Protection

Insect control: Flea beetles, cutworms, lygus bugs, cabbage seedpod weevil, diamondback moth, bertha armyworm, cabbage looper and occasionally imported cabbageworm, grasshoppers and slugs might require control. Seed treatments are available for flea beetle and cutworm control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓		✓	✓			✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Oriental Mustard

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,389.00
Estimated Farm Gate Price ($/lb.) (B)		0.30
Estimated Gross Revenue ($/ac.) (AxB)=(C)		416.70
Expenses Per Acre		
Variable Expenses/Acre		
Seed		20.10
- Seed Treatments/Inoculants		0.54
Fertilizer - Nitrogen		25.35
- Phosphorus (P_2O_5)		13.29
- Sulphur and Other		4.79
Plant Protection - Herbicides		42.58
- Insecticides		8.12
- Fungicides		0.00
Machinery Operating - Fuel		19.03
- Repair		9.00
Custom Work and Hired Labour		20.30
Crop Insurance Premium		5.87
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		3.39
Total Variable Expenses (D)		**175.59**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**270.93**
Return Per Acre		
Return Over Variable Expenses (C-D)		241.11
Return Over Total Expenses (C-G)		145.77
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		585.29
To Cover Total Expenses		903.10
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.13
To Cover Total Expenses		0.20

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	948.00
Return Over Variable Expenses	108.81
Return Over Total Expenses	13.47

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 6 lb./ac. is used. The brown soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 53 lb./ac. N, 28 lb./ac. P_2O_5 and 15 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure and pathogen levels in the field.

Crop Protection

Insect control: Flea beetles, cutworms, lygus bugs, cabbage seedpod weevil, diamondback moth, bertha armyworm, cabbage looper and occasionally imported cabbageworm, grasshoppers and slugs might require control. Seed treatments are available for flea beetle and cutworm control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Mustards are relatively resilient to weed competition. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓		✓	✓			✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Yellow Mustard

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,168.00
Estimated Farm Gate Price ($/lb.) (B)		0.42
Estimated Gross Revenue ($/ac.) (AxB)=(C)		490.56
Expenses Per Acre		
Variable Expenses/Acre		
Seed		33.50
- Seed Treatments/Inoculants		0.90
Fertilizer - Nitrogen		21.53
- Phosphorus (P_2O_5)		11.87
- Sulphur and Other		4.79
Plant Protection - Herbicides		42.24
- Insecticides		0.00
- Fungicides		0.00
Machinery Operating - Fuel		19.03
- Repair		9.00
Custom Work and Hired Labour		20.30
Crop Insurance Premium		8.47
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		3.44
Total Variable Expenses (D)		**178.29**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**273.63**
Return Per Acre		
Return Over Variable Expenses (C-D)		312.27
Return Over Total Expenses (C-G)		216.93
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		424.49
To Cover Total Expenses		651.50
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.15
To Cover Total Expenses		0.23

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	661.40
Return Over Variable Expenses	99.50
Return Over Total Expenses	4.16

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 10 lb./ac. is used. The brown soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 45 lb./ac. N, 25 lb./ac. P_2O_5 and 15 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure and pathogen levels in the field.

Crop Protection

Insect control: Flea beetles, cutworms, lygus bugs, cabbage seedpod weevil, diamondback moth, bertha armyworm, cabbage looper and occasionally imported cabbageworm, grasshoppers and slugs might require control. Seed treatments are available for flea beetle and cutworm control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Yellow mustard has limited herbicide options. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓		✓	✓			✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Sunflower Oilseed (EMSS)

Economics

	My Farm	Dark Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		2,293.00
Estimated Farm Gate Price ($/lb.) (B)		0.25
Estimated Gross Revenue $/ac.) (AxB)=(C)		573.25
Expenses Per Acre		
Variable Expenses/Acre		
Seed		55.12
- Seed Treatments/Inoculants		0.00
Fertilizer - Nitrogen		35.88
- Phosphorus (P_2O_5)		14.24
- Sulphur and Other		26.07
Plant Protection - Herbicides		56.14
- Insecticides		30.69
- Fungicides		0.00
Machinery Operating - Fuel		19.93
- Repair		10.15
Custom Work and Hired Labour		20.80
Crop Insurance Premium		7.43
Utilities and Miscellaneous		4.23
Interest on Variable Expenses		5.52
Total Variable Expenses (D)		**286.21**
Other Expenses/Acre		
Building Repair		0.65
Property Taxes		5.19
Business Overhead		3.48
Machinery Depreciation		41.78
Building Depreciation		1.40
Machinery Investment		16.11
Building Investment		0.61
Land Investment		40.31
Total Other Expenses (E)		**109.53**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**395.74**
Return Per Acre		
Return Over Variable Expenses (C-D)		287.04
Return Over Total Expenses (C-G)		177.51
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		1,144.83
To Cover Total Expenses		1,582.95
Break Even Price ($/lb.).		
To Cover Variable Expenses		0.12
To Cover Total Expenses		0.17

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	1,763.70
Return Over Variable Expenses	154.72
Return Over Total Expenses	45.19

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: This guide assumes a producer will seed to achieve a plant population of 26,000/ac. This estimation is for the moist long season area with both dark brown and black soil, located in the south-east of the province.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 75 lb./ac. N, 30 lb./ac. P_2O_5, 50 lb./ac. K_2O and 26 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: If a sunflower midge infestation is anticipated, new fields should be established away from fields damaged the previous season. Crop rotation can be used to reduce insects and disease pressure and pathogen levels in the field.

Crop Protection

Insect control: Wireworms, cutworms, sunflower beetle, grasshoppers, lygus bugs, sunflower seed weevil, banded sunflower moth and sunflower moth might require control. Seed treatments are available for wireworm and sunflower beetle control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓		✓	✓	✓		✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Soybean**

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		32.0	34.5	37.5
Estimated Farm Gate Price ($/bu.) (B)		10.93	10.93	10.93
Estimated Gross Revenue ($/ac.) (AxB)=(C)		349.76	377.09	409.88
Expenses Per Acre				
Variable Expenses/Acre				
Seed		98.00	98.00	98.00
- Seed Treatments/Inoculants		14.00	14.00	14.00
Fertilizer - Nitrogen		2.44	2.63	2.82
- Phosphorus (P_2O_5)		11.39	12.34	13.29
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		54.82	54.82	54.82
- Insecticides		2.17	2.17	2.17
- Fungicides		0.00	0.00	0.00
Machinery Operating - Fuel		13.77	17.21	21.52
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		22.30	22.05	23.05
Crop Insurance Premium		5.68	4.53	3.69
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		4.66	4.76	4.91
Total Variable Expenses (D)		**241.46**	**246.90**	**254.65**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**336.80**	**356.43**	**374.22**
Return Per Acre				
Return Over Variable Expenses (C-D)		108.30	130.19	155.23
Return Over Total Expenses (C-G)		12.96	20.66	35.66
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		22.09	22.59	23.30
To Cover Total Expenses		30.81	32.61	34.24
Break Even Price ($/bu.)				
To Cover Variable Expenses		7.55	7.16	6.79
To Cover Total Expenses		10.53	10.33	9.98

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	23.88	24.62	28.29
Return Over Variable Expenses	19.55	22.20	54.56
Return Over Total Expenses	-75.79	-87.33	-65.01

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A plant population of four to five plants per square foot is recommended. This corresponds to 150,000 to 200,000 plants per acre. Seed survivability averages 75 per cent, which is usually achieved when using a drill. Solid seeded soybeans with narrow rows (eight to 10 in.) improve crop yields, raise the height of bottom pods and reduce the need for multiple in-crop herbicide applications for weed control.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 6 lb./ac. N and 28 lb./ac. P2O5 for the black soil zone, 6 lb./ac. N and 26 lb./ac. P2O5 for the dark brown soil zone and 5 lb./ac. N and 24 lb./ac. P2O5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests. Soybean requires a specific species of rhizobia not native to Saskatchewan soil. Double inoculation is recommended on new fields. Most varieties come pre-treated and pre-liquid inoculated. Addition of a second inoculant of granular or peat is recommended.

Crop Rotation: Crop rotation will help to reduce disease pressure by reducing or maintaining low pathogen levels in the field. Soybeans are not competitive with weeds.

Crop Protection

Insect control: Wireworms, seedcorn maggot, cutworms, soybean aphid, leafhoppers, lygus bugs, spider mites, armyworms, corn earworm and grasshoppers might require control. Seed treatments are available for wireworm and seedcorn maggot control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Soybeans need to be kept free of weeds from the first trifoliate leaf to the third trifoliate leaf to minimize yield losses. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
		✓	✓	✓	✓	✓		✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

**These soybeans are glyphosate tolerant.

2021 Desi Chickpea

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		2,138.00
Estimated Farm Gate Price ($/lb.) (B)		0.30
Estimated Gross Revenue ($/ac.) (AxB)=(C)		641.40
Expenses Per Acre		
Variable Expenses/Acre		
Seed		39.06
- Seed Treatments/Inoculants		13.81
Fertilizer - Nitrogen		3.35
- Phosphorus (P_2O_5)		17.09
- Sulphur and Other		0.00
Plant Protection - Herbicides		59.72
- Insecticides		20.84
- Fungicides		31.58
Machinery Operating - Fuel		19.93
- Repair		9.00
Custom Work and Hired Labour		19.80
Crop Insurance Premium		12.25
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		4.91
Total Variable Expenses (D)		**254.57**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**349.91**
Return Per Acre		
Return Over Variable Expenses (C-D)		386.83
Return Over Total Expenses (C-G)		291.49
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		848.56
To Cover Total Expenses		1,166.36
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.12
To Cover Total Expenses		0.16

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	1,146.40
Return Over Variable Expenses	89.35
Return Over Total Expenses	-5.99

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: Seed rates for the Desi Chickpea is 93 lb./ac. The brown soil zone is recommended.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility cost is based on nutrient removal rates given the targeted crop yield. These are 7 lb./ac. N and 36 lb./ac. P_2O_5. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Diverse crop rotations are recommended to help reduce disease pressure and suppress weed growth.

Crop Protection

Insect control: Wireworms, cutworms, pea aphid, potato leafhopper and grasshoppers might require control. Seed treatments are available for wireworm control.

Disease control: Ascochyta blight is a foliar disease that can result in yield losses when environmental conditions favour disease development. When disease pressure is high more than one fungicide application may be required. Two fungicide applications are included in this estimate. Fungicide application decisions should be based on disease risk during the growing season.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓	✓	✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Kabuli Chickpea, Large

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		2,491.00
Estimated Farm Gate Price ($/lb.) (B)		0.30
Estimated Gross Revenue ($/ac.) (AxB)=(C)		747.30
Expenses Per Acre		
Variable Expenses/Acre		
Seed		60.90
- Seed Treatments/Inoculants		13.81
Fertilizer - Nitrogen		3.83
- Phosphorus (P_2O_5)		18.51
- Sulphur and Other		0.00
Plant Protection - Herbicides		59.72
- Insecticides		20.84
- Fungicides		31.58
Machinery Operating - Fuel		19.93
- Repair		9.00
Custom Work and Hired Labour		19.80
Crop Insurance Premium		11.64
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		5.36
Total Variable Expenses (D)		**278.15**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**373.49**
Return Per Acre		
Return Over Variable Expenses (C-D)		469.15
Return Over Total Expenses (C-G)		373.81
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		927.17
To Cover Total Expenses		1,244.97
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.11
To Cover Total Expenses		0.15

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	1,477.10
Return Over Variable Expenses	164.98
Return Over Total Expenses	69.64

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 145 lb./ac. is used. The brown soil zone is recommended.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility cost is based on nutrient removal rates given the targeted crop yield. These are 8 lb./ac. N and 39 lb./ac. P2O5. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure.

Crop Protection

Insect control: Wireworms, cutworms, pea aphid, potato leafhopper and grasshoppers might require control. Seed treatments are available for wireworm control.

Disease control: Ascochyta blight is a foliar disease that can result in yield losses when environmental conditions favor disease development. When disease pressure is high more than once fungicide application may be required. Two fungicide applications are included in this estimate. Fungicide application decisions should be based on disease risk during the growing season.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓	✓	✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Kabuli Chickpea, Small

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		2,161.00
Estimated Farm Gate Price ($/lb.) (B)		0.25
Estimated Gross Revenue ($/ac.) (AxB)=(C)		540.25
Expenses Per Acre		
Variable Expenses/Acre		
Seed		45.78
- Seed Treatments/Inoculants		13.81
Fertilizer - Nitrogen		3.35
- Phosphorus (P_2O_5)		15.19
- Sulphur and Other		0.00
Plant Protection - Herbicides		59.72
- Insecticides		20.84
- Fungicides		31.58
Machinery Operating - Fuel		19.93
- Repair		9.00
Custom Work and Hired Labour		19.80
Crop Insurance Premium		11.14
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		4.98
Total Variable Expenses (D)		**258.35**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**353.69**
Return Per Acre		
Return Over Variable Expenses (C-D)		281.90
Return Over Total Expenses (C-G)		186.56
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		1,033.40
To Cover Total Expenses		1,414.76
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.12
To Cover Total Expenses		0.16

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	1,234.60
Return Over Variable Expenses	50.30
Return Over Total Expenses	-45.04

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 109 lb./ac. is used. The brown soil zone is recommended.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility cost is based on nutrient removal rates given the targeted crop yield. These are 7 lb./ac. N and 32 lb./ac. P_2O_5. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure.

Crop Protection

Insect control: Wireworms, cutworms, pea aphid, potato leafhopper and grasshoppers might require control. Seed treatments are available for wireworm control.

Disease control: Ascochyta blight is a foliar disease that can result in yield losses when environmental conditions favour disease development. When disease pressure is high, more than one fungicide application may be required. Two fungicide applications are included in this estimate. Fungicide application decisions should be based on disease risk during the growing season.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓	✓	✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Large Green Lentils

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (lb./ac.) (A)		1,676.00	1,852.00	1,918.00
Estimated Farm Gate Price ($/lb.) (B)		0.27	0.27	0.27
Estimated Gross Revenue ($/ac.) (AxB)=(C)		452.51	500.03	517.85
Expenses Per Acre				
Variable Expenses/Acre				
Seed		41.86	41.86	41.86
- Seed Treatments/Inoculants		13.01	13.01	13.01
Fertilizer - Nitrogen		2.13	2.10	2.10
- Phosphorus (P_2O_5)		9.02	9.97	9.97
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		62.60	62.60	60.86
- Insecticides		8.77	8.77	8.77
- Fungicides		19.50	19.50	19.50
Machinery Operating - Fuel		13.77	17.21	21.52
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		21.80	21.05	22.80
Crop Insurance Premium		9.86	10.52	12.03
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		4.22	4.35	4.50
Total Variable Expenses (D)		**218.77**	**225.34**	**233.30**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**314.11**	**334.87**	**352.87**
Return Per Acre				
Return Over Variable Expenses (C-D)		233.74	274.69	284.55
Return Over Total Expenses (C-G)		138.40	165.16	164.98
Break Even Yield (lbs./ac.)				
To Cover Variable Expenses		810.26	834.59	864.08
To Cover Total Expenses		1163.38	1240.27	1306.96
Break Even Price ($/lb.)				
To Cover Variable Expenses		0.13	0.12	0.12
To Cover Total Expenses		0.19	0.18	0.18

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (lb./ac.)	1080.30	1234.60	1146.40
Return Over Variable Expenses	72.91	108.00	76.22
Return Over Total Expenses	-22.43	-1.53	-43.35

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seed rate of 91 lb./ac. is used for all soil zones.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 4 lb./ac. N and 21 lb./ac. P_2O_5 for the black soil zone, 4 lb./ac. N and 21 lb./ac. P_2O_5 for the dark brown soil zone and 4 lb./ac. N and 19 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation will help to reduce disease pressure by reducing pathogen levels in the field. Extended rotations are essential in fields with risk of aphanomyces root rot. Lentils are not competitive with weeds.

Crop Protection

Insect control: Wireworms, cutworms lygus bugs, potato leafhopper, pea aphid and grasshoppers might require control. Seed treatments are available for wireworm control.

Disease control: Fungicides will offer protection against foliar diseases such as anthracnose. Fungicide application should be based on disease risk. This estimation includes the cost of a single fungicide application. In years with high disease pressure and extended periods of favorable environmental conditions, more than one fungicide application may be required.

Weed control: Lentils need to be kept weed-free until the 10 node stage to minimize yield losses. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓	✓	✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Red Lentils

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (lb./ac.) (A)		1,940.00	2,205.00	2,491.00
Estimated Farm Gate Price ($/lb.) (B)		0.25	0.25	0.25
Estimated Gross Revenue ($/ac.) (AxB)=(C)		485.00	551.25	622.75
Expenses Per Acre				
Variable Expenses/Acre				
Seed		22.80	22.80	22.80
- Seed Treatments/Inoculants		17.49	17.49	17.49
Fertilizer - Nitrogen		2.25	2.44	2.82
- Phosphorus (P_2O_5)		10.44	11.39	13.29
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		62.60	62.60	60.86
- Insecticides		8.77	8.77	8.77
- Fungicides		19.50	19.50	19.50
Machinery Operating - Fuel		13.77	17.21	21.52
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		21.80	21.05	22.80
Crop Insurance Premium		9.30	9.47	8.21
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		3.95	4.07	4.22
Total Variable Expenses (D)		**204.90**	**211.18**	**218.66**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**300.24**	**320.71**	**338.23**
Return Per Acre				
Return Over Variable Expenses (C-D)		280.10	340.07	404.09
Return Over Total Expenses (C-G)		184.76	230.54	284.52
Break Even Yield (lbs./ac.)				
To Cover Variable Expenses		819.60	844.72	874.62
To Cover Total Expenses		1,200.97	1,282.85	1,352.92
Break Even Price ($/lb.).				
To Cover Variable Expenses		0.11	0.10	0.09
To Cover Total Expenses		0.15	0.15	0.14

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (lb./ac.)	1,697.60	1,433.00	1,477.10
Return Over Variable Expenses	219.50	147.07	150.62
Return Over Total Expenses	124.16	37.54	31.05

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seed rate of 60 lb./ac. is used for all soil zones.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 6 lb./ac. N and 28 lb./ac. P_2O_5 for the black soil zone, 5 lb./ac. N and 24 lb./ac. P_2O_5 for the dark brown soil zone and 5 lb./ac. N and 22 lb./ac. P_2O_5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation will help to reduce disease pressure by reducing pathogen levels in the field. Extended rotations are essential in fields with risk of aphanomyces root rot. Lentils are not competitive with weeds.

Crop Protection

Insect control: Wireworms, cutworms, lygus bugs, potato leafhopper, pea aphid and grasshoppers might require control. Seed treatments are available for wireworm control.

Disease control: Fungicides will offer protection against foliar diseases such as anthracnose. Fungicide application should be based on disease risk. This estimation includes the cost of a single fungicide application. In years with high disease pressure and extended periods of favorable environmental conditions, more than one fungicide application may be required.

Weed control: Lentils need to be kept weed free until the 10 node stage to minimize yield losses. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓	✓	✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Edible Green Peas

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		43.4	52.9	61.4
Estimated Farm Gate Price ($/bu.) (B)		9.00	9.00	9.00
Estimated Gross Revenue ($/ac.) (AxB)=(C)		390.60	476.10	552.60
Expenses Per Acre				
Variable Expenses/Acre				
Seed		37.68	42.90	48.40
- Seed Treatments/Inoculants		10.88	12.39	13.98
Fertilizer - Nitrogen		3.35	4.07	4.74
- Phosphorus (P_2O_5)		15.66	18.99	22.31
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		61.63	61.63	45.08
- Insecticides		8.77	8.77	8.77
- Fungicides		28.13	28.13	28.13
Machinery Operating - Fuel		13.77	17.21	21.52
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.30	20.30	20.30
Crop Insurance Premium		4.80	4.48	5.23
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		4.27	4.59	4.62
Total Variable Expenses (D)		**221.47**	**237.85**	**239.45**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**316.81**	**347.38**	**359.02**
Return Per Acre				
Return Over Variable Expenses (C-D)		169.13	238.25	313.15
Return Over Total Expenses (C-G)		73.79	128.72	193.58
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		24.61	26.43	26.61
To Cover Total Expenses		35.20	38.60	39.89
Break Even Price ($/bu.)				
To Cover Variable Expenses		5.10	4.50	3.90
To Cover Total Expenses		7.30	6.57	5.85

Yield Sensitivity (same expenses, but average yield)			
Provincial Average Yield (bu./ac.)	33.80	35.27	37.85
Return Over Variable Expenses	82.73	79.58	101.20
Return Over Total Expenses	-12.61	-29.95	-18.37

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production. Note that the yield data used in this calculation combines both yellow and green peas. Producers are reminded to insert their own target yields.

Seeding: A seed rate of 176 lb./ac. is used in the black soil zone, 156 lb/ac in the dark brown soil zone and 137 lb./ac. in the brown soil zone.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 10 lb./ac. N and 47 lb./ac. P2O5 for the black soil zone, 9 lb./ac. N and 40 lb./ac. P2O5 for the dark brown soil zone and 7 lb./ac. N and 33 lb./ac. P2O5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation will help to reduce pea leaf weevil and disease pressure by reducing pathogen levels in the field. Extended rotations are essential in fields with risk of aphanomyces root rot.

Crop Protection

Insect control: Wireworms, cutworms, lygus bugs, leafhoppers, pea aphid, grasshoppers, alfalfa looper and pea leaf weevil might require control. Seed treatments are available for wireworm and pea leaf weevil control. Refer to the ministry's forecasts of local pea leaf weevil pressures at saskatchewan.ca/crops.

Disease control: Fungicides will offer protection against foliar diseases such as mycosphaerella blight. This estimation includes the cost of a single fungicide application. Fungicide application should be based on disease risk.

Weed control: Weeds in peas need to be controlled until 10 to 14 days after emergence to minimize yield losses. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓		✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Edible Yellow Peas

Economics

	My Farm	Brown	Dark Brown	Black
Revenue Per Acre				
Target Yield (bu./ac.) (A)		43.4	52.9	61.4
Estimated Farm Gate Price ($/bu.) (B)		7.50	7.50	7.50
Estimated Gross Revenue ($/ac.) (AxB)=(C)		325.50	396.75	460.50
Expenses Per Acre				
Variable Expenses/Acre				
Seed		31.62	36.20	40.54
- Seed Treatments/Inoculants		10.96	12.55	14.14
Fertilizer - Nitrogen		3.35	4.07	4.74
- Phosphorus (P_2O_5)		15.66	18.99	22.31
- Sulphur and Other		0.00	0.00	0.00
Plant Protection - Herbicides		61.63	61.63	45.08
- Insecticides		8.77	8.77	8.77
- Fungicides		28.13	28.13	28.13
Machinery Operating - Fuel		13.77	17.21	21.52
- Repair		9.00	10.15	11.49
Custom Work and Hired Labour		20.30	20.30	20.30
Crop Insurance Premium		4.80	4.48	5.23
Utilities and Miscellaneous		3.23	4.23	4.88
Interest on Variable Expenses		4.15	4.46	4.47
Total Variable Expenses (D)		**215.38**	**231.18**	**231.60**
Other Expenses/Acre				
Building Repair		0.49	0.65	0.88
Property Taxes		3.97	5.19	7.87
Business Overhead		2.28	3.48	4.08
Machinery Depreciation		37.05	41.78	47.28
Building Depreciation		1.05	1.40	1.90
Machinery Investment		14.28	16.11	18.23
Building Investment		0.46	0.61	0.83
Land Investment		35.77	40.31	38.50
Total Other Expenses (E)		**95.34**	**109.53**	**119.57**
Labour and Management* (F)				
Total Expenses (D+E+F)=(G)		**310.72**	**340.71**	**351.18**
Return Per Acre				
Return Over Variable Expenses (C-D)		110.12	165.57	228.90
Return Over Total Expenses (C-G)		14.78	56.04	109.32
Break Even Yield (bu./ac.)				
To Cover Variable Expenses		28.72	30.82	30.88
To Cover Total Expenses		41.43	45.43	46.82
Break Even Price ($/bu.)				
To Cover Variable Expenses		4.96	4.37	3.77
To Cover Total Expenses		7.16	6.44	5.72
Yield Sensitivity (same expenses, but average yield)				
Provincial Average Yield (bu./ac.)		33.80	35.27	37.85
Return Over Variable Expenses		38.12	33.35	52.28
Return Over Total Expenses		-57.22	-76.18	-67.30

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production. Note that the yield data used in this calculation combines both yellow and green peas. Producers are reminded to insert their own target yields.

Seeding: A seed rate of 178 lb./ac. is used in the black soil zone, 158 lb./ac. in the dark brown soil zone and 138 lb./ac. in the brown soil zone.

Fertilization: Inoculant with the correct strains of rhizobium should be applied. Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 10 lb./ac. N and 47 lb./ac. P2O5 for the black soil zone, 9 lb./ac. N and 40 lb./ac. P2O5 for the dark brown soil zone and 7 lb./ac. N and 33 lb./ac. P2O5 for the brown soil zone. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation will help to reduce pea leaf weevil and disease pressure by reducing pathogen levels in the field. Extended rotations are essential in fields with risk of aphanomyces root rot.

Crop Protection

Insect control: Wireworms, cutworms, lygus bugs, leafhoppers, pea aphid, grasshoppers, alfalfa looper and pea leaf weevil might require control. Seed treatments are available for wireworm and pea leaf weevil control. Refer to the ministry's forecasts of local pea leaf weevil pressures at saskatchewan.ca/crops.

Disease control: Fungicides will offer protection against foliar diseases, such as mycosphaerella blight. This estimation includes the cost of a single fungicide application. Fungicide application should be based on disease risk.

Weed control: Weeds in peas need to be controlled until 10 to 14 days after emergence to minimize yield losses. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓	✓		✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Black Bean

Economics

	My Farm	Dark Brown
Revenue Per Acre		
Target Yield (lb./ac.) (A)		1,200.00
Est. Farm Gate Price $/lb. (B)		0.33
Estimated Gross Revenue/ac (AxB)=C		396.00
Expenses Per Acre		
Variable Expenses/Acre		
Seed		74.25
- Seed Treatments/Inoculants		10.89
Fertilizer - Nitrogen		9.57
- Phosphorus (P_2O_5)		3.32
- Sulphur and Other		0.00
Plant Protection - Herbicides		56.33
- Insecticides		62.72
- Fungicides		9.04
Machinery Operating - Fuel		17.21
- Repair		10.15
Custom Work and Hired Labour		19.80
Crop Insurance Premium		13.40
Utilities and Miscellaneous		4.23
Interest on Variable Expenses		5.72
Total Variable Expenses (D)		**296.64**
Other Expenses/Acre		
Building Repair		0.65
Property Taxes		5.19
Business Overhead		3.48
Machinery Depreciation		41.78
Building Depreciation		1.40
Machinery Investment		16.11
Building Investment		0.61
Land Investment		40.31
Total Other Expenses (E)		**109.53**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**406.17**
Return Per Acre		
Return Over Variable Expenses (C-D)		99.36
Return Over Total Expenses (C-G)		-10.17
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		898.91
To Cover Total Expenses		1,230.82
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.25
To Cover Total Expenses		0.34

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	870.00
Return Over Variable Expenses	-9.54
Return Over Total Expenses	-119.07

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 55 lb./ac. is used. The dark brown soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 20 lb./ac. N and 7 lb./ac. P_2O_5. Producers are encouraged to use their own rates based on soil tests. Black beans do not respond well to inoculant.

Crop Rotation: Crop rotation can be used to reduce disease pressure and pathogen levels in the field.

Crop Protection

Insect control: Aphids, leafhoppers, cutworms, corn borer and lygus bugs might require control.

Disease control: White mould and common bacterial blight are the most common diseases of black bean. This estimate includes a single fungicide application for white mould management.

Weed control: Black beans need to be maintained weed free between the second trifoliate stage and the onset of flowering (three weeks to five or six weeks after emergence) to minimize yield loss. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓		✓	✓			✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Faba bean

Economics

	My Farm	Black
Revenue Per Acre		
Target Yield (lb./ac.) (A)		3,902.00
Estimated Farm Gate Price ($/lb.) (B)		0.12
Estimated Gross Revenue ($/ac.) (AxB)=(C)		468.24
Expenses Per Acre		
Variable Expenses/Acre		
Seed		42.23
- Seed Treatments/Inoculants		0.00
Fertilizer - Nitrogen		5.17
- Phosphorus (P_2O_5)		24.21
- Sulphur and Other		0.00
Plant Protection - Herbicides		58.54
- Insecticides		20.84
- Fungicides		28.13
Machinery Operating - Fuel		19.93
- Repair		11.49
Custom Work and Hired Labour		19.80
Crop Insurance Premium		10.81
Utilities and Miscellaneous		4.88
Interest on Variable Expenses		4.84
Total Variable Expenses (D)		**250.88**
Other Expenses/Acre		
Building Repair		0.88
Property Taxes		7.87
Business Overhead		4.08
Machinery Depreciation		47.28
Building Depreciation		1.90
Machinery Investment		18.23
Building Investment		0.83
Land Investment		38.50
Total Other Expenses (E)		**119.57**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**370.45**
Return Per Acre		
Return Over Variable Expenses (C-D)		217.36
Return Over Total Expenses (C-G)		97.79
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		2,090.63
To Cover Total Expenses		3,087.08
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.06
To Cover Total Expenses		0.09
Yield Sensitivity (same expenses, but average yield)		
Provincial Average Yield (lb./ac.)		2,976.20
Return Over Variable Expenses		106.26
Return Over Total Expenses		-13.31

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seed rate of 181 lb./ac. is used. Faba beans are recommended for the black soil zone.

Fertilization: Inoculant with correct strain of rhizobium should be applied. Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 18.4 lb./ac and 87 lb./ac. P2O5. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure by allowing infected crop residue to decompose between susceptible crops.

Crop Protection

Insect control: Wireworms, cutworms, lygus bugs, potato leafhopper, pea aphid, grasshoppers, and pea leaf weevil might require control. Seed treatments are available for wireworm and pea leaf weevil control. Refer to ministry forecasts of local pea leaf weevil pressures at saskatchewan.ca/crops

Disease control: Chocolate spot is a foliar disease that can result in poor seed set and flower abortion. A single application of fungicides for the management of chocolate spot has been included in this estimate. Fungicide application should be based on disease risk within the growing season.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓			✓

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

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2021 Camelina

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,852.00
Estimated Farm Gate Price ($/lb.) (B)		0.22
Estimated Gross Revenue ($/ac.) (AxB)=(C)		407.44
Expenses Per Acre		
Variable Expenses/Acre		
Seed		21.00
- Seed Treatments/Inoculants		0.00
Fertilizer - Nitrogen		40.66
- Phosphorus (P_2O_5)		9.49
- Sulphur and Other		4.79
Plant Protection - Herbicides		9.60
- Insecticides		0.00
- Fungicides		0.00
Machinery Operating - Fuel		19.03
- Repair		9.00
Custom Work and Hired Labour		20.30
Crop Insurance Premium		4.88
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		2.79
Total Variable Expenses (D)		**144.77**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**240.11**
Return Per Acre		
Return Over Variable Expenses (C-D)		262.67
Return Over Total Expenses (C-G)		167.33
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		658.04
To Cover Total Expenses		1091.41
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.08
To Cover Total Expenses		0.13

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	1014.10
Return Over Variable Expenses	78.33
Return Over Total Expenses	-17.01

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seed rate of 6 lb./ac. is used. Camelina is commonly grown in the brown soil zone.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 85 lb./ac. N, 20 lb./ac. P_2O_5 and 15 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure by allowing infected crop residue to decomposse between susceptible crops.

Crop Protection

Insect control: Only one biological insecticide, with efficacy against bertha armyworm and diamondback moth, is registered for this crop.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Camelina has no registered herbicide options other than a pre-harvest application of glyphosate in the previous crop to manage perrenial weeds as well as a glyphosate burn off prior to seeding.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓				✓				

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Canary seed

Economics

	My Farm	Dark Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,918.00
Estimated Farm Gate Price ($/lb.) (B)		0.28
Estimated Gross Revenue ($/ac.) (AxB)=(C)		537.04
Expenses Per Acre		
Variable Expenses/Acre		
Seed		18.20
- Seed Treatments/Inoculants		0.00
Fertilizer - Nitrogen		30.14
- Phosphorus (P_2O_5)		25.63
- Sulphur and Other		17.16
Plant Protection - Herbicides		42.15
- Insecticides		14.66
- Fungicides		7.15
Machinery Operating - Fuel		18.12
- Repair		10.15
Custom Work and Hired Labour		22.05
Crop Insurance Premium		7.35
Utilities and Miscellaneous		4.23
Interest on Variable Expenses		4.27
Total Variable Expenses (D)		**221.26**
Other Expenses/Acre		
Building Repair		0.65
Property Taxes		5.19
Business Overhead		3.48
Machinery Depreciation		41.78
Building Depreciation		1.40
Machinery Investment		16.11
Building Investment		0.61
Land Investment		40.31
Total Other Expenses (E)		**109.53**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**330.79**
Return Per Acre		
Return Over Variable Expenses (C-D)		315.78
Return Over Total Expenses (C-G)		206.25
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		790.22
To Cover Total Expenses		1181.40
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.12
To Cover Total Expenses		0.17

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	1212.50
Return Over Variable Expenses	118.24
Return Over Total Expenses	8.71

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column. Calculations provided are for the 80th percentile of production.

Seeding: A seeding rate of 35 lb./ac. is assumed. Canary seed is grown in all soil zones, but heavy clay soil is preferred.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 63 lb./ac. N, 54 lb./ac. P_2O_5, 33 lb./ac. K_2O and 17 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure by allowing infected crop residue to decompose between susceptible crops.

Crop Protection

Insect control: Canary seed is susceptible to aphids, so an insecticide application to manage for aphids is assumed.

Disease control: Septoria leaf mottle is a foliar disease of canary seed that can result in yield losses when environmental conditions favor disease development. A single fungicide application for septoria leaf mottle has been included in this estimate. Fungicide application decisions should be made based on disease risk during the growing season.

Weed control: Canary seed has limited herbicide options for grass control making application of a soil-active for wild oat control a necessity. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓	✓	✓	✓			

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Caraway (Second Season)*

Economics

	My Farm	Black
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		850.00
Estimated Farm Gate Price ($/lb.) (B)		0.50
Estimated Gross Revenue ($/ac.) (AxB)=(C)		425.00
Expenses Per Acre		
Variable Expenses/Acre		
Seed		5.40
- Seed Treatments/Inoculants		0.63
Fertilizer - Nitrogen		30.62
- Phosphorus (P_2O_5)		23.73
- Sulphur and Other		10.31
Plant Protection - Herbicides		32.58
- Insecticides		0.00
- Fungicides		0.00
Machinery Operating - Fuel		36.24
- Repair		22.98
Custom Work and Hired Labour		19.80
Crop Insurance Premium		17.69
Utilities and Miscellaneous		4.88
Interest on Variable Expenses		4.03
Total Variable Expenses (D)		**208.89**
Other Expenses/Acre (for two years since biennial)		
Building Repair		1.76
Property Taxes		15.74
Business Overhead		8.17
Machinery Depreciation		94.57
Building Depreciation		3.80
Machinery Investment		19.89
Building Investment		1.66
Land Investment		76.99
Total Other Expenses (E)		**222.58****
Labour and Management*** (F)		
Total Expenses (D+E+F)=(G)		**431.47**
Return Per Acre (after two years)		
Return Over Variable Expenses (C-D)		216.11
Return Over Total Expenses (C-G)		-6.47
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		417.78
To Cover Total Expenses		862.93
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.25
To Cover Total Expenses		0.51

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	529.10
Return Over Variable Expenses	55.66
Return Over Total Expenses	-166.92

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: Seed rate of 12 lb./ac. is used. Caraway is primarily a biennial crop and typically seeded with a companion crop so that the field provides a return in both crop years. Caraway and Coriander have the same management cost. Therefore, if no companion crop is sown, include production costs used for coriander, with the exception of seeding, harvest and handling, to account for management costs of caraway in the first year. Recommended soil is black.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 64 lb./ac. N, 50 lb./ac. P_2O_5 and 29 lb./ac. K_2O. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure by allowing infected crop residue to decompose between susceptible crops. Crops in the carrot family are very sensitive to Group 2 residues in the soil. Please refer to the ministry's Guide to Crop Protection available at saskatchewan.ca/crops for more information.

Crop Protection

Insect control: A limited number of registered products is available for slugs in field crops.

Disease control: The cost of fungicide is not included in this estimate. Blossom blight can result in yield losses when conditions favor disease development.

Weed control: Herbicide costs below are only for the second year of production. If caraway is not companion cropped in the first year, it requires the same weed control as coriander. Refer to the coriander section for more information. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used (Second Year Only)								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
					✓	✓		

* These expenses are for the second year of production only. The first year is identical to coriander.

**These other costs are carried for two years because caraway is a biennial crop. If a cover crop is grown and harvested in the first year, these costs could be halved.

***Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Coriander (and Caraway First Season)

Economics

	My Farm	Dark Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,565.00
Estimated Farm Gate Price ($/lb.) (B)		0.35
Estimated Gross Revenue ($/ac.) (AxB)=(C)		547.75
Expenses Per Acre		
Variable Expenses/Acre		
Seed		11.00
- Seed Treatments/Inoculants		1.31
Fertilizer - Nitrogen		39.23
- Phosphorus (P_2O_5)		21.84
- Sulphur and Other		0.00
Plant Protection - Herbicides		63.17
- Insecticides		0.00
- Fungicides		34.99
Machinery Operating - Fuel		18.12
- Repair		10.15
Custom Work and Hired Labour		19.80
Crop Insurance Premium		11.09
Utilities and Miscellaneous		4.23
Interest on Variable Expenses		4.62
Total Variable Expenses (D)		**239.55**
Other Expenses/Acre		
Building Repair		0.65
Property Taxes		5.19
Business Overhead		3.48
Machinery Depreciation		41.78
Building Depreciation		1.40
Machinery Investment		16.11
Building Investment		0.61
Land Investment		40.31
Total Other Expenses (E)		**109.53**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**349.08**
Return Per Acre		
Return Over Variable Expenses (C-D)		308.20
Return Over Total Expenses (C-G)		198.67
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		684.42
To Cover Total Expenses		997.37
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.15
To Cover Total Expenses		0.22

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	903.90
Return Over Variable Expenses	76.82
Return Over Total Expenses	-32.71

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 25 lb./ac. is used. The dark brown soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 82 lb./ac. N and 46 lb./ac. P2O5. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure. Crops in the carrot family are very sensitive to Group 2 residues in the soil. Please refer to the Ministry's Guide to Crop Protection available at saskatchewan.ca/crops for more information.

Crop Protection

Insect control: A limited number of insecticides are registered for aphid and slug control.

Disease control: Blossom blight can result in yield losses when conditions favour disease development. A single application of fungicide for blossom blight management is included in this estimate. Fungicide applications should be made based on disease risk during the growing season.

Weed control: Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓		✓	✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Fenugreek

Economics

	My Farm	Brown
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,175.00
Estimated Farm Gate Price ($/lb.) (B)		0.26
Estimated Gross Revenue ($/ac.) (AxB)=(C)		305.50
Expenses Per Acre		
Variable Expenses/Acre		
Seed		15.00
- Seed Treatments/Inoculants		0.00
Fertilizer - Nitrogen		1.44
- Phosphorus (P_2O_5)		6.65
- Sulphur and Other		0.00
Plant Protection - Herbicides		25.68
- Insecticides		0.00
- Fungicides		0.00
Machinery Operating - Fuel		15.40
- Repair		9.00
Custom Work and Hired Labour		19.80
Crop Insurance Premium		0.00
Utilities and Miscellaneous		3.23
Interest on Variable Expenses		1.89
Total Variable Expenses (D)		**98.08**
Other Expenses/Acre		
Building Repair		0.49
Property Taxes		3.97
Business Overhead		2.28
Machinery Depreciation		37.05
Building Depreciation		1.05
Machinery Investment		14.28
Building Investment		0.46
Land Investment		35.77
Total Other Expenses (E)		**95.34**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**193.43**
Return Per Acre		
Return Over Variable Expenses (C-D)		207.42
Return Over Total Expenses (C-G)		112.07
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		377.25
To Cover Total Expenses		743.94
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.08
To Cover Total Expenses		0.16

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	740.30
Return Over Variable Expenses	94.40
Return Over Total Expenses	-0.95

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 30 lb./ac. is used. The brown soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 3 lb./ac. N and 14 lb./ac. P_2O_5. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure and pathogen levels in the field.

Crop Protection

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

Weed control: Fenugreek has limited herbicide options. Herbicide costs are based on the following herbicide timings. Please refer to general assumptions for details.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓			✓	✓		

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

2021 Quinoa

Economics

	My Farm	Black
Revenue Per Acre		
Target Yield (lbs./ac.) (A)		1,000.00
Estimated Farm Gate Price ($/lb.) (B)		0.65
Estimated Gross Revenue ($/ac.) (AxB)=(C)		650.00
Expenses Per Acre		
Variable Expenses/Acre		
Seed		63.00
- Seed Treatments/Inoculants		0.50
Fertilizer - Nitrogen		20.09
- Phosphorus (P_2O_5)		8.54
- Sulphur and Other		5.82
Plant Protection - Herbicides		9.60
- Insecticides		18.82
- Fungicides		0.00
Machinery Operating - Fuel		15.40
- Repair		11.49
Custom Work and Hired Labour		19.80
Crop Insurance Premium		20.93
Utilities and Miscellaneous		4.88
Interest on Variable Expenses		3.91
Total Variable Expenses (D)		**202.80**
Other Expenses/Acre		
Building Repair		0.88
Property Taxes		7.87
Business Overhead		4.08
Machinery Depreciation		47.28
Building Depreciation		1.90
Machinery Investment		18.23
Building Investment		0.83
Land Investment		38.50
Total Other Expenses (E)		**119.57**
Labour and Management* (F)		
Total Expenses (D+E+F)=(G)		**322.37**
Return Per Acre		
Return Over Variable Expenses (C-D)		447.20
Return Over Total Expenses (C-G)		327.63
Break Even Yield (lbs./ac.)		
To Cover Variable Expenses		312.00
To Cover Total Expenses		495.96
Break Even Price ($/lb.)		
To Cover Variable Expenses		0.20
To Cover Total Expenses		0.32

Yield Sensitivity (same expenses, but average yield)	
Provincial Average Yield (lb./ac.)	630.00
Return Over Variable Expenses	206.70
Return Over Total Expenses	87.13

Agronomics

Note: refer to the online calculator on saskatchewan.ca/crops for the "My Farm" column.

Seeding: A seed rate of 10 lb./ac. is used. The black soil zone is recommended.

Fertilization: Fertility costs are based on nutrient removal rates given the targeted crop yield. These are: 42 lb./ac. N, 18 lb./ac. P_2O_5, 11 lb./ac. K_2O and 6 lb./ac. S. Producers are encouraged to use their own rates based on soil tests.

Crop Rotation: Crop rotation can be used to reduce disease pressure and pathogen levels in the field.

Crop Protection

Insect control: A limited number of insecticides are registered for European corn borer control.

Disease control: The cost of fungicide is not included in this estimate. Disease risk assessment is encouraged to guide fungicide application decisions.

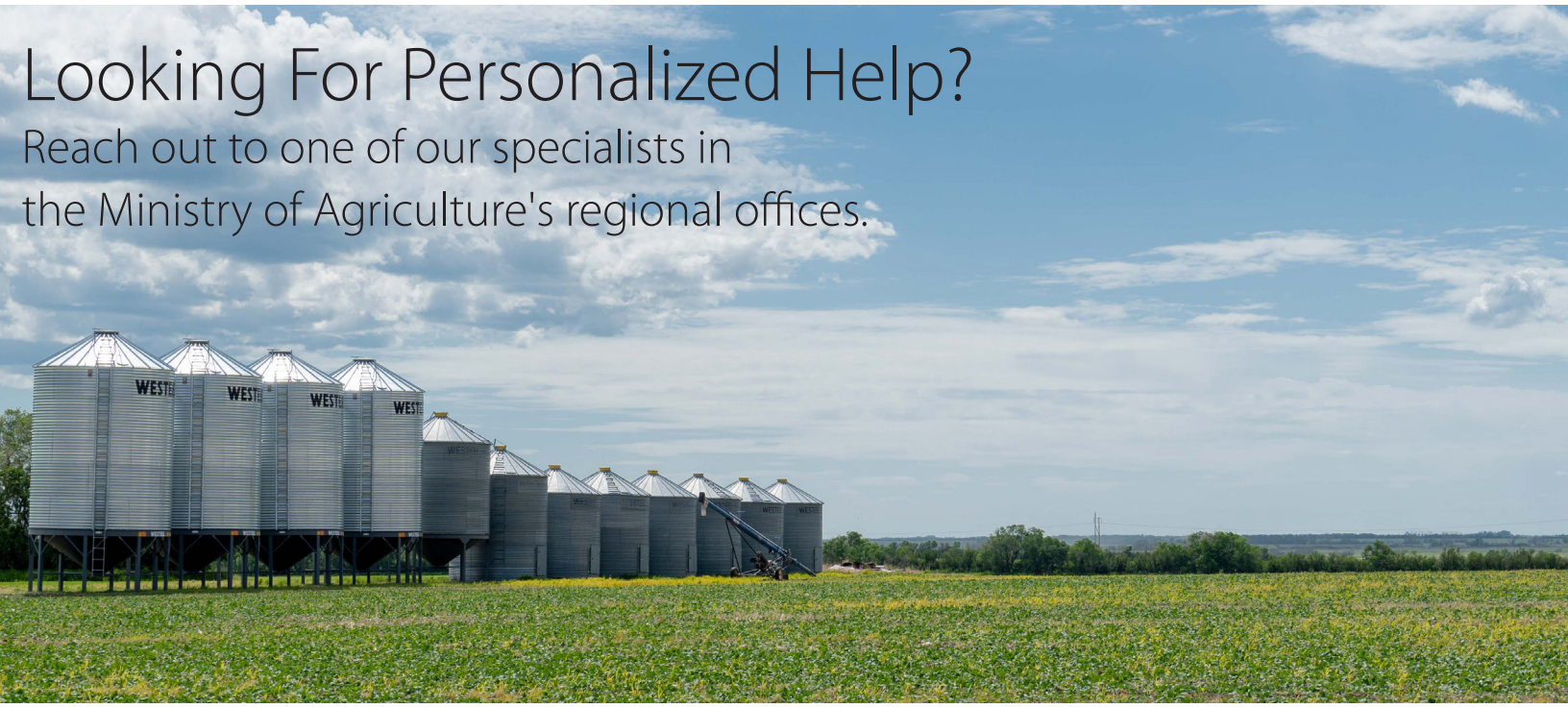
Weed control: Quinoa has no registered herbicide options. However, producers can apply a pre-harvest application of glyphosate in the previous crop to manage perennial weeds, as well as a glyphosate burn off prior to seeding. Please see below chart and refer to general assumptions for details on pre-harvest glyphosate application.

Application Timing Window Used								
Pre-harvest	Fall-applied	Pre-seed		Soil	In-crop			Desiccation
		1	2		1	2	3	
✓		✓						

*Farm managers need to determine their own actual labour and management costs and add it to total expenses.

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